Product datasheet

Vertical lifting fabric door
Megadoor VL3190

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in
door opening solutions



Copyright and Disclaimer Notice

Although the contents of this publication have been compiled with the greatest possible care, ASSA ABLOY Entrance Systems cannot accept liability for any damage that might arise from errors or omissions in this publication. We also reserve the right to make appropriate technical modifications/replacements without prior notice.

No rights can be derived from the contents of this document.

Color guides: Color differences may occur due to different printing and publication methods.

ASSA ABLOY, Besam, Crawford, Megadoor and Albany, as words and logos, are examples of trademarks owned by ASSA ABLOY Entrance Systems or companies within the ASSA ABLOY Group.

Copyright © ASSA ABLOY Entrance Systems AB 2006-2014.

No part of this publication may be copied or published by means of scanning, printing, photocopying, microfilm or any other process whatsoever without prior permission in writing by ASSA ABLOY Entrance Systems.

All rights reserved.

Technical facts

Features

Max size: (W/H) depending on wind load*	19000 x 20000 mm
Door leaf thickness:	290 mm
Fabric types:	Standard: Polyester (coating: plasticised PVC) Options: Arctic, sound reduction, heat resistant, security
Color:	9 standard colours
Guide rails material:	Aluminium
Windows:	Vision panels (width 800 mm standard)
Seals:	Bottom, side and top seal
Operation:	Standard: Electrical operator Optional: Automated operation, Access control, Safety functions

*Other sizes available on request.

Note! For larger openings, see Megadoor Special Doors with virtually no size limitations other than what is practical. Megadoor Special Doors can be delivered as large single belt doors (with 2-motor drive), large single wire rope doors or multiple door systems.

Performance

Operating speed:	0.15-0.25 m/s
Wind load resistance: (differential pressure)	Can withstand almost any wind load by varying the size and the spacing of the intermediate sections.
Wind speed, door in motion:	< 20 m/s
Sound reduction (standard):	15 dB Rw (ISO 717)
Water resistance:	Class 3 (EN 12425, 0.11 kPa (for a closed door)
Air permeability:	Class 2-3 (EN 12426)
Operating environment temperature range:	-35 °C to +70 °C
Thermal transmittance:	Depending on door size. Specific data on request.

Contents

Copyright and Disclaimer Notice 2

Technical facts 3

Contents 4

1. Description 6

1.1 General 6
1.2 Standard 6
1.3 Options 6
1.4 Door leaf 7
1.4.1 Construction 7
1.4.2 Intermediate section 7
1.4.3 Bottom section 7
1.4.4 Safety arresters 7
1.4.5 Wind locking 7
1.4.6 Door leaf material 8
1.4.7 Colors 8
1.4.8 Options 8
1.5 Guide rails 9
1.6 Header box 9
1.6.1 Enclosing the header box 10
1.7 Operating system 11
1.7.1 Electrical operation 11
1.7.2 Lifting belts 11
1.7.3 Drive unit 11
1.7.4 Control unit 12
1.7.5 Access and automation 13

2. Specifications 14

2.1 Clear width and clear height 14
2.2 Performance 14
2.3 Environmental tolerance 14
2.4 Surface treatment 14
2.5 Door leaf 15
2.5.1 Fabric data 15
2.6 Operating system 19
2.6.1 General specifications 19

3. CEN Performance 20

3.1 Lifetime expectation 20
3.2 Resistance to windload 20
3.3 Resistance to water penetration 20
3.4 Air permeability 20
3.5 Thermal transmittance 21
3.6 Acoustic insulation 21
3.7 Operating forces and safe openings 21

4. Building preparations 22

4.1 Installation 22
4.1.1 Installation of the header box 22
4.1.2 Mounting surface for header box 23
4.2 Mounting surface for guide rails 24
4.2.1 Mounting alternatives for guide rails 24
4.3 Installation of the guide rails 25
4.3.1 Installation of the control unit 25

5. Space requirements 26

5.1 Space requirements for header box 26
5.2 Space requirements for operation 27
5.3 Space requirements for control unit 28
5.4 Space requirements for maintenance 28

6. Service 29

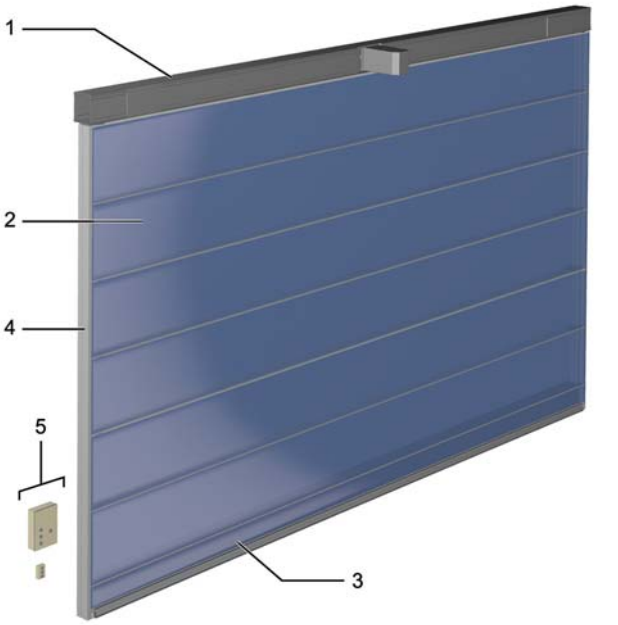
Index 30

1. Description

1.1 General

The Megadoor VL3190 vertical lifting fabric door is the preferred door model for the extremely large door openings needed for aircraft hangars and shipyard halls, it serves equally well in any door application where innovative design will make a positive difference to the construction cost and subsequent operating efficiency.

The unique design and structure offers durability, tightness, energy efficiency, operational reliability and minimum maintenance. Every door is individually designed to meet application requirements, for example wind load.



The Megadoor VL3190 vertical lifting fabric door has five main components:

1) Header box
2) Door leaf
3) Bottom section
4) Guide rails
5) Control cabinet

1.2 Standard

The Megadoor VL3190 vertical lifting fabric door is supplied with the following specifications as standard:

Door leaf:	Polyester, 1100 dtex with plasticised PVC coating
Safety:	Safety arresters
Operation:	Drive unit + control unit
Colors:	Choice of 9 standard colours

1.3 Options

Megadoor provides a wide range of options and accessories to customise the Megadoor VL3190 vertical lifting fabric door to any customer's needs. For example:

Door leaf:	Arctic, heat resistant, sound reduction and security fabrics Vision panels Clamp strip covers
Header box:	Protective cladding
Colors:	Optional colors on request
Operation:	Automation

1.4 Door leaf

1.4.1 Construction

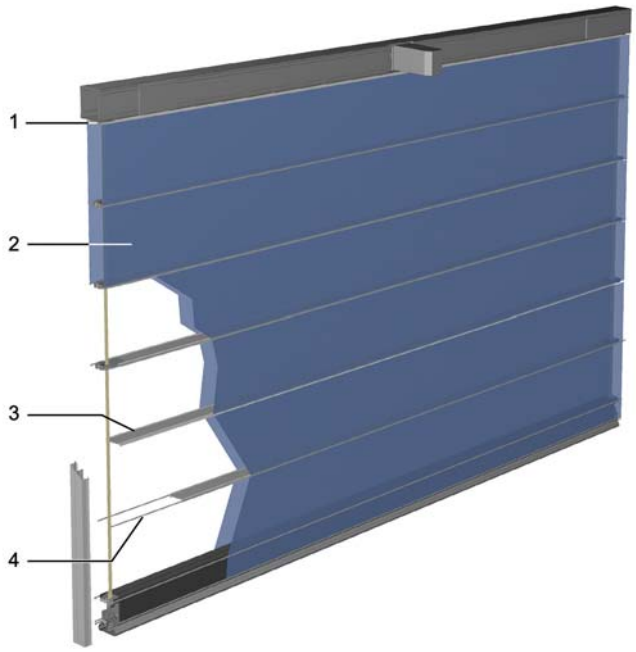
The door leaf is made of two layers of very strong vinyl-coated polyester fabric, separated by aluminium intermediate sections. The aluminium top section is bolted to the header box, the steel and aluminium bottom section is connected to the lifting belt via the safety arresters.

The fabric is attached to both sides of the intermediate sections, top section and bottom section with self-tapping screws through aluminium clamp strips, providing maximum tightness.

Wind load is transferred to the vertical guide rails by the horizontal aluminium sections of the door leaf.

1.4.2 Intermediate section

The intermediate sections, which are made of extruded aluminium, are fitted at each end with lubrication-free guide blocks, which travel in the guide rails on each side of the door leaf. The section depth is 290 mm for the door with belt operation.



1) Header box
2) Guide rail
3) Safety arrester
4) Bottom section

1.4.3 Bottom section

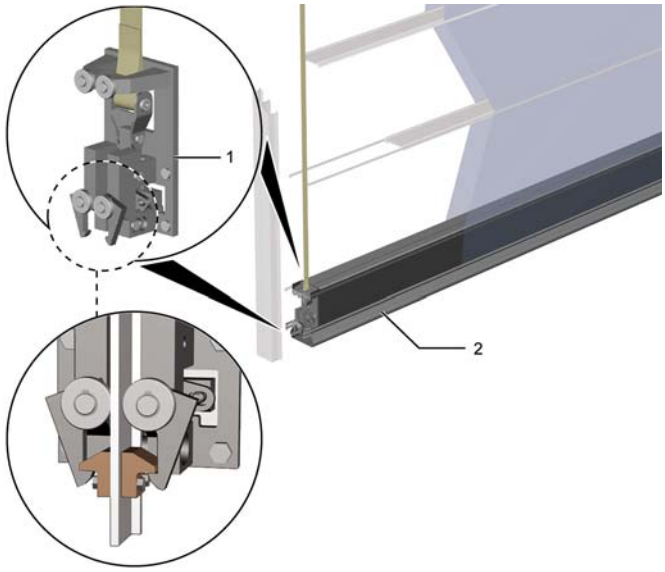
The bottom section is made of steel. A rubber seal fixed to the section ensures tightness against the floor/ground.

1.4.4 Safety arresters

The lifting belts are attached to the patented safety arresters, which in turn are fixed to the bottom section. In the unlikely event of a belt failure, the safety arresters are activated and immediately lock the door in the guide rails.

The safety function is tested and certified by TÜV.

Wind locking



Bottom section:

1) Safety arrester
2) Rubber seal

1.4.5 Wind locking

Strong winds subject a large door to a substantial load. Megadoor safety arresters therefore have a unique built-in wind locking, which is activated and locks the bottom beam when the door is closed.

1.4.6 Door leaf material

Standard Fabric

The standard door-leaf fabric is a single sheet of heavy-duty vinyl-coated polyester. The fabric is resistant to mechanical abrasion and sparks generated from mechanical processes such as welding.

The standard fabric is available in 9 standard colours, however other colors are available on request.

Arctic Fabric

The arctic fabric replaces the standard fabric in environments where the temperature can be as low as -54°C.

Sound-reduction Fabric

The sound-reduction fabric is for use in environments where the transmission of sound through the door must be reduced. It is installed on both sides of the door leaf behind the standard fabric.

Heat-resistant Fabric

The heat-resistant fabric replaces the standard fabric on the inside of the door leaf when there is a requirement to contain heat and/or chemical hazards. It is available with three different coatings dependant on the environment where it is going to be used.

Security Fabric

The security fabric is for use in environments where security is important. It is similar to the standard fabric with the addition of galvanized steel wires inside the fabric. It is installed on both sides of the door leaf behind the standard fabric to a height of about two meters.

Vision Panels

Vision panels (windows) are available for the standard and arctic fabrics to improve light admission and visibility through the door leaf.

1.4.7 Colors

The RAL-colors are as close as possible to the official RAL HR collection.

1.4.7.1 Standard colors

	RAL 1001
	RAL 3001
	RAL 5005
	RAL 6009
	RAL 7004
	RAL 7016
	RAL 9006
	RAL 9016
	Translucent white

1.4.7.2 Optional colors

Other colors are available on request.

1.4.8 Options

Clamp strip covers

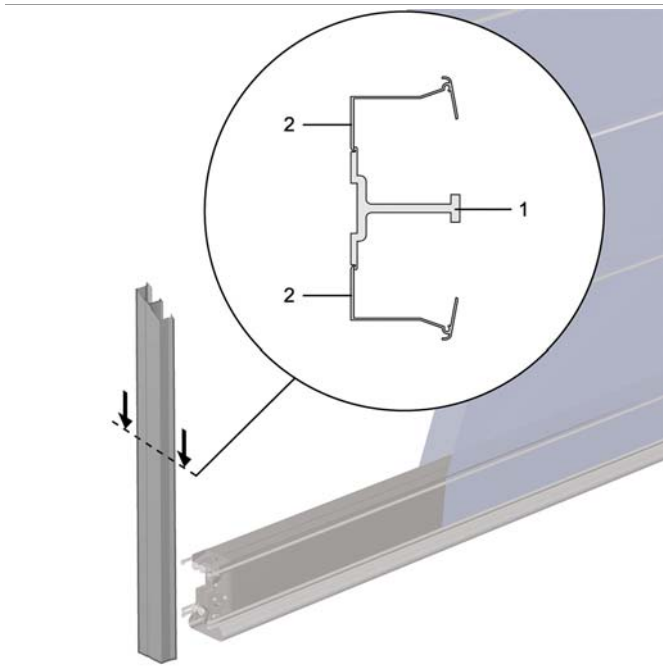
Clamp strip covers are plastic strips that clip onto the clamp strips. They are available in the same standard colors as the fabric.

The benefits of the clamp strip covers are:

- Improve the appearance of the door leaf
- Cover the screws
- Protect the door leaf from discoloration in certain environments.

1.5 Guide rails

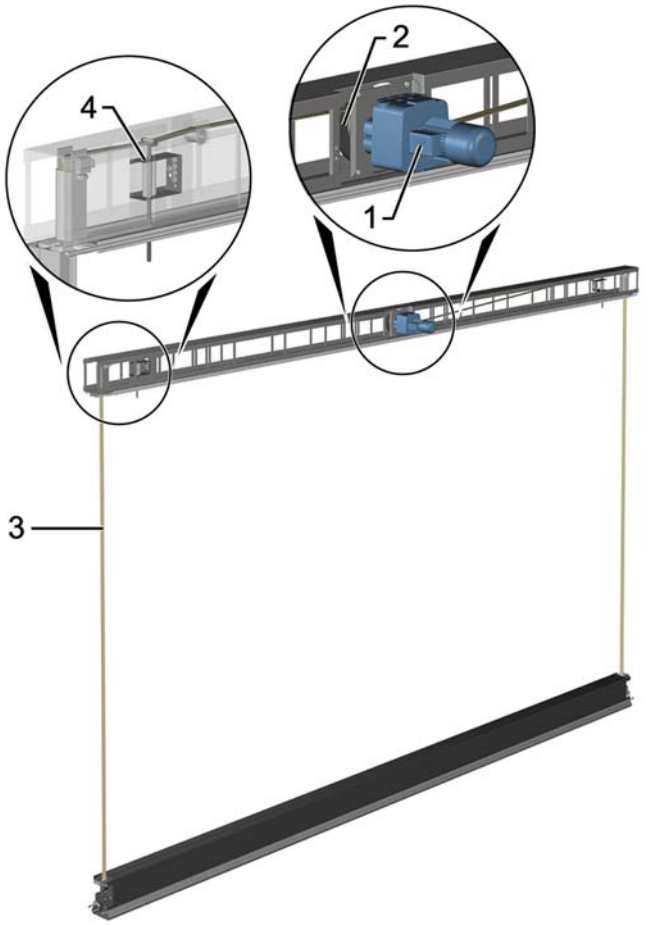
The extruded guide rails are made up of three parts, a rail surrounded by two outer sections. The guide blocks in the intermediate sections travel along the guide rail and guide the door. The design of the guide rails ensures that air leaks are minimised.



1) Guide rail
2) Seal angle

1.6 Header box

The door leaf with the bottom section is suspended by a firm box structure of steel, which contains the drive unit and limit switch units with position sensors and devices for checking belt status.

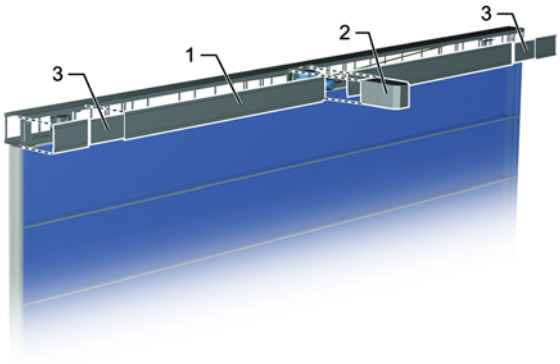


1) Drive unit
2) Belt drum
3) Lifting belt
4) Safety boxes

1.6.1 Enclosing the header box

1.6.1.1 Enclosed motor side

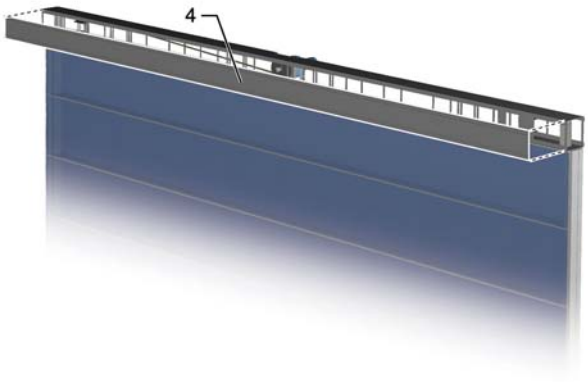
The door is delivered as standard with the motor side enclosed. Components that need to be accessible for inspection are located at the ends of the header box behind hatches. The rest of the header box has a removable sheet-steel enclosure between the hatches.



1) Enclosed motor side (standard)
2) Protective casing (extra)
3) Inspection hatches (standard)

1.6.1.2 Enclosed "non-motor side" (extra)

For fitting in a door opening, with the drive unit facing inwards, the “non-motor side” should be fitted with a fixed sheet-metal cover. Inspection is done from the motor side.



4) Enclosed "non-motor side" (extra)

1.6.1.3 Protective casing over motor (extra)

For outdoor fitting or in dirty environments, the motor should be fully protected. The protective casing is made of powder coated steel sheet. The casing is provided with a hatch to facilitate easy access to the motor for possible manual emergency operation if required. The entire cover can be removed.

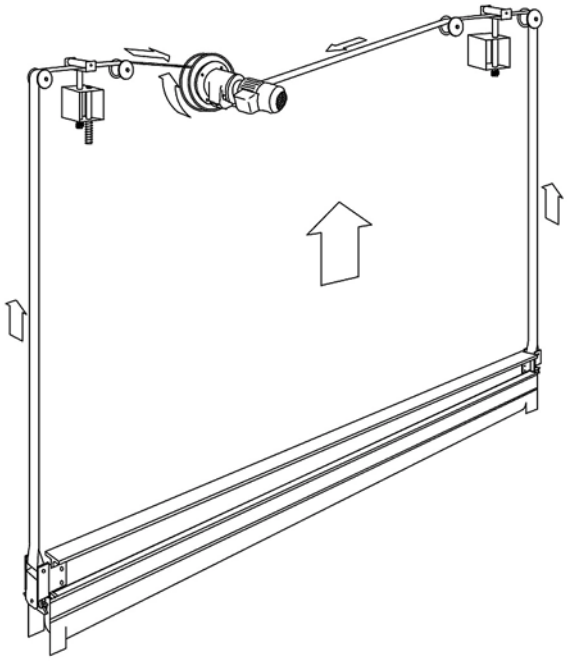
1.7 Operating system

1.7.1 Electrical operation

The Megadoor VL3190 vertical lifting fabric door is always supplied with an electrical operating system, a control unit near the door and a gear motor in the header box.

The door is opened by an impulse from the UP-button.

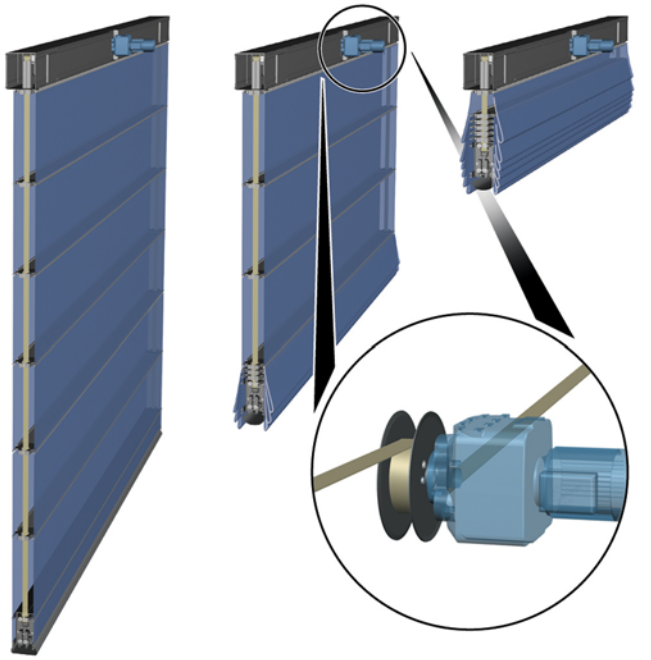
The door is closed by pressing the DOWN-button continuously (Hold to run).



1.7.2 Lifting belts

The bottom section is lifted using belts, which are wound up on the belt drum. The belts are fitted with sewn loops for attachment to the safety arresters. The belts are not sensitive to rust, dirt and dust, and are tested and certified.

Fully closed **Fully open**



1.7.3 Drive unit

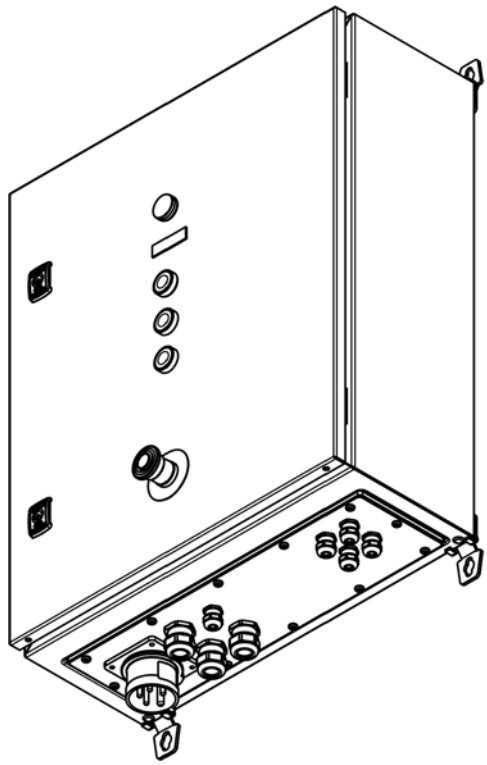
The gear motor, which is equipped with a brake, also has a hand-operated brake release and crank, so that the door can be opened or closed in the event of a power failure. The belt drum is directly mounted with a keyed joint on the output shaft of the gear motor.

1.7.4 Control unit

The door is supplied with a PLC-based control unit installed near the door. The control unit commands the gear motor via push buttons.

The UP button opens the door by impulse signal. The DOWN button is set to hold-to-run. The gear motor can be disabled from the control unit for emergency hand-crank operation by switching off the mains.

The control unit is available in a standard model and an extended model. By default the standard model supports the most basic functions whereas the extended model supports all available functions.



1.7.4.1 PLC

The control unit contains a PLC and an LCD with integrated buttons to navigate through the screens for information or to configure the door operation. The PLC is programmed with factory default settings before delivery. The following information is given:

- Number of days of operation and number of door openings from the start since the door was last serviced.
- Current settings
- Alarm codes
- Control unit temperature (option)

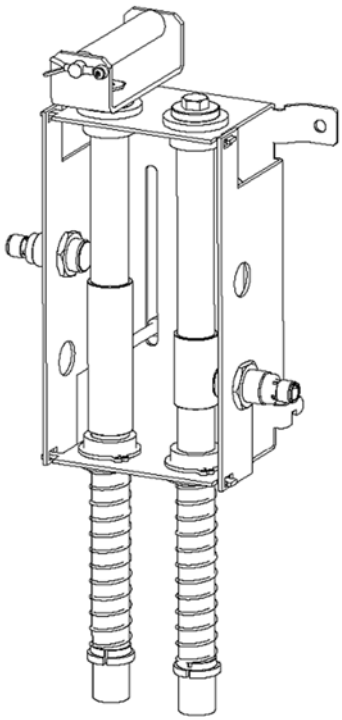
The standard control unit does not include an LCD and does not support displaying of information or door operation configuration.

1.7.4.2 Temperature control

As an option, the control unit can be equipped with temperature control devices such as a fan or a heating element.

1.7.4.3 Safety boxes

The safety boxes are low in maintenance and have a high ingress protection level (IP67) and temperature tolerance. The inductive proximity switches monitor belt rupture and door over travel.



1.7.5 Access and automation

The standard control unit supports external push button box and one safety photocell.

1.7.5.1 Additional automatic functions

Megadoor offers a wide range of functions that allow advanced opening and safety control.

1.7.5.2 Control functions

Free contacts

Potential free switching contacts are available on blocks in the control cabinet, from the functions "door open", and "door closed". These functions can be used to connect signal devices, air curtains, airlock function, etc.

Reduced opening



When it is unnecessary or undesirable to fully open a door, an absolute encoder is used to configure a reduced opening position.

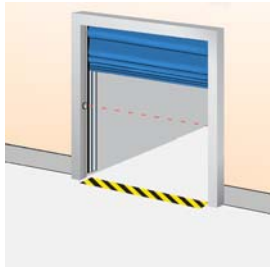
External push button box



An extra control box is installed outside the building or inside close to the door if the main control unit needs to be installed away from the door opening.
Installed on the inside or outside wall beside the door.

1.7.5.3 Safety functions

Safety photocells 1-channel



A set of a photocell transmitter with reflector or receiver is installed in the door opening. If the photocell beam is interrupted during closing, the door will stop in less than 30mm and reverse to the fully open position.
Installed in the door opening.

Warning lights - Orange flashing lights



Flashing light during door movement.
Duration of start warning is configurable.
May be combined with or replaced by sounder.
Installed on the inside and-or outside wall beside the door.

Emergency power switch



A power switch can be enabled, as a backup system, in case of a power failure.
Supplied with power inlet socket.

2. Specifications

2.1 Clear width and clear height

The standard Megadoor VL3190 vertical lifting fabric door is delivered in the following size range:

Max size: (WxH) depending on wind load.*	19000 x 20000 mm

*Other sizes available on request.

Note! For larger openings, see Megadoor Special Doors with virtually no size limitations other than what is practical.

Megadoor Special Doors can be delivered as large single belt doors (with 2-motor drive), large single wire rope doors or multiple door systems.

2.2 Performance

Operating speed:	0.15-0.25 m/s
Wind load resistance: (differential pressure)	Can withstand almost any wind load by varying the size and the spacing of the intermediate sections.
Wind speed, door in motion:	< 20 m/s
Sound reduction (standard):	15 dB Rw (ISO 717)
Water resistance:	Class 3 (EN 12425, 0.11 kPa (for a closed door)
Air permeability:	Class 2-3 (EN 12426)
Operating environment temperature range:	-35 °C to +70 °C
Thermal transmittance:	Depending on door size. Specific data on request.

2.3 Environmental tolerance

Heat and cold resistance	-35°C to +70°C
Atmospheric humidity	below dew point
Presence of particles	< 1000 µg/m³ air
Mechanical load, blasting	Not directly aimed.
Differential pressure, closed door	Class 3 (EN12424, temporary 0.7 kPa)
Wind speed, in motion	< 20 m/s
Acidity	Condensate at 5<pH<9
Explosive fumes or dust	No occurrence.

*In the normal version, the door is suited for operation in environments within the limits stated above. If the requirements exceed these limits (e.g. wind load), the door can often be modified on request.

2.4 Surface treatment

Steel components	For corrosion, category 3 according to ISO 12944.2. Higher class on request.
Other parts	Aluminium, plastic, stainless steel, zinc electroplated steel (~ 10µ). Fixing elements are mainly hot dip galvanized (FZV).
	Door leaf screws are corrosion protected with Geomet.

2.5 Door leaf

2.5.1 Fabric data

2.5.1.1 Standard fabric

Application	Standard		
Use	Standard		
Coating	Plasticized PVC		
Fabric	Polyester, 1100 dtex		
Weight	700 g/m²		
Heat- and cold resistance	-35°C to +70°C. DIN EN 1876-2 1998-01. (-30°C to + 70°C for the Translucent white fabric)		
Tensile strength	Warp : 2500N/5 cm acc. DIN 53354, EN ISO 1421 Weft : 2000N/5 cm acc. DIN 53354, EN ISO 1421		
Tear resistance	Warp : 400N acc DIN 53363 Weft : 300N acc. DIN 53363		
Resistance to light	7 - 8 (on a scale 0-8). ISO 105-B02 1998		
UV-stabilized	Yes		
Fire behavior	M2 (NF P 92 507 2004)		
Reaction to fire	B - s2,d0 (EN 13501-1 2007)		
Mildew resistant	Yes		
Rot resistant	Yes		
Radar reflection	0.3 dB, - 0.1%		
Lacquered	Yes		
Standard colors	• Tan • Red • Blue • Green • Grey • Anthracite grey • White • White aluminium • Translucent white	NCS 2010Y-40R NCS 2070-R NCS S3560-R80B NCS 8010-G10Y NCS 3500 NCS 8005-B20G NCS 0500	RAL 1001 RAL 3001 RAL 5005 RAL 6009 RAL 7004 RAL 7016 RAL 9016 RAL 9006
Logotype	Optional		
Vision panels	Optional		

2.5.1.2 Arctic fabric

Application	Environmental temperatures down to -54°C		
Use	Replaces standard fabric		
Coating	Plasticized PVC		
Fabric	Polyester, 1100 dtex		
Weight	700 g/m²		
Heat- and cold resistance	-54°C to +70°C. DIN EN 1876-2 1998-01		
Tensile strength	Warp : 2500N/5 cm acc. DIN 53354, EN ISO 1421 Weft : 2000N/5 cm acc. DIN 53354, EN ISO 1421		
Tear resistance	Warp : 400N acc DIN 53363 Weft : 300N acc. DIN 53363		
Resistance to light	7 - 8 (on a scale 0-8). ISO 105-B02		
UV-stabilized	Yes		
Fire behavior	M2 (NF P 92 507 2004)		
Reaction to fire	B - s2,d0 (EN 13501-1 2007)		
Mildew resistant	Yes		
Rot resistant	Yes		
Radar reflection	0.3 dB, - 0.1%		
Lacquered	Yes		
Standard colors	• Tan • Red • Blue • Green • Grey • Anthracite grey • White • White aluminium	NCS 2010Y-40R NCS 2070-R NCS S3560-R80B NCS 8010-G10Y NCS 3500 NCS 8005-B20G NCS 0500	RAL 1001 RAL 3001 RAL 5005 RAL 6009 RAL 7004 RAL 7016 RAL 9016 RAL 9006
Logotype	Optional		

Note! Not in combination with:

- Vision panels
- Sound reduction fabric
- Heat resistant fabric
- Security fabric

2.5.1.3 Sound reduction fabric

Application	Sound reduction
Use	On both sides of the door behind the standard fabric
Coating	Plasticized PVC
Fabric	Polyester, 1100 dtex
Weight	1850 g/m²
Sound reduction (incl. standard fabric)	Index Rw23dB*, tested by SP Swedish National Testing and research Institute
Heat- and cold resistance	-30°C to +70°C, acc. SFS-EN 1876-1
Tensile strength	Warp: 3000N/5 cm acc. DIN 53354 Weft: 2900N/5 cm acc. DIN 53354
Tear resistance	Warp: 380N acc DIN 53356 Weft: 300N acc. DIN 53356
Flame resistant	Acc. SIS 650082, DIN 4102-B1
Comments	Space for fabric folding must be increased by 100 mm on each side of the door, to avoid fabric wear.

Note! Must always be quoted by ASSA ABLOY Entrance Systems.

* Weighted apparent sound reduction index acc. ISO 717-1. For more information, ask for SP-report P103341, dated 15 June 2001 'Determination of sound insulation of an industrial door according to SS-EN ISO-140-3:95'.

2.5.1.4 Heat resistant fabric - Silicone coating

Application	• Hot air environment • Coating highly resistant to chemicals • Dirt and oil repellent • Electrical insulation • Weatherproof, UV and oxidation resistant
Use	Replacing standard fabric
Designation	Alpha Maritex 3200-2-SS
Coating	Silicon rubber on both sides
Fabric	Woven glass fibre EC9-136
Weight	555 g/m²
Heat- and cold resistance	Coating -36°C to +260°C
Tensile strength	Warp : 450N/ cm acc. DIN ISO 4606 Weft : 440N/cm acc. DIN ISO 4606
Flame resistance	Acc. BS 476:Part 7, 1971 Part 6, 1989, M0. BS6853:1987 App. B, IMO resolution A653 (16)
Approvals	• Lloyds: SVG/F92/110, SAS F970017 • Powergen 08/65/242, LUL E 1042 A3 • National Power 08/GS/259
Comments	• Never combine standard and heat resistant fabric (for example upper part of the door with standard and lower part of the door with heat resistant fabric). • Protect the bottom sealing with the fabric as well. • When the door is installed against a wall on the cool side, the folding space on the hot side must be increased by at least 100 mm to avoid fabric wear. • The motor should be placed on the cool side.A heat radiation shield below the motor may be necessary. • All cables must be protected. • The clear height should be as large as possible.

Note! Must always be quoted by ASSA ABLOY Entrance Systems.

2.5.1.5 Heat resistant fabric - Aluminium coating

Application	Hot air and high radiation temperatures inside (e.g. foundries). Good heat reflection properties.
Use	On the inside of the door (never on the outside) replacing standard fabric.
Designation	332 AL-HT
Coating	Aluminium pigments on polyurethane adhesive on one side of the fabric.
Fabric	E-glass EC9-136 (cross twill)
Weight	490 g/m²
Heat- and cold resistance	From contact coating +200°C (not continuously)
Tensile strength	Warp : 800N/cm acc. DIN 53857 T1 Weft : 500N/cm acc. DIN 53857 T1
Comments	• Never combine standard and heat resistant fabric (for example upper part of the door with standard and lower part of the door with heat resistant fabric). • Protect the bottom sealing with the fabric as well. • When the door is installed against a wall on the cool side, the folding space on the hot side must be increased by at least 100 mm to avoid fabric wear. • The motor should be placed on the cool side.A heat radiation shield below the motor may be necessary. • All cables must be protected. • The clear height should be as large as possible.

Note! Must always be quoted by ASSA ABLOY Entrance Systems.

2.5.1.6 Heat resistant fabric - Aluminium polyurethane coating

Application	Fire barrier
Use	On the inside of the door (never on the outside) replacing standard fabric.
Designation	W2167 Gp2
Coating	Two sides aluminium grey polyurethane
Thickness	0.8 mm
Fabric	Woven glass fibre, Atlas 1/8
Weight	690 g/m²
Heat resistance	+450°C
Tensile strength	Warp : 1350N/cm acc. EN ISO 13934-1 Weft : 1260N/cm acc. EN ISO 13934-1
Fire classification	Incombustible according to M0 (French standard)
Comments	• Never combine standard and heat resistant fabric (for example upper part of the door with standard and lower part of the door with heat resistant fabric). • Protect the bottom sealing with the fabric as well. • When the door is installed against a wall on the cool side, the folding space on the hot side must be increased by at least 100 mm to avoid fabric wear. • The motor should be placed on the cool side.A heat radiation shield below the motor may be necessary. • All cables must be protected. The clear height should be as large as possible.

Note! Must always be quoted by ASSA ABLOY Entrance Systems.

2.5.1.7 Security fabric

Application	Protection against burglary
Use	On both sides of the door, behind the standard fabric. Up to approximately 2 meters from the floor
Designation	Protector PRO
Fabric	PVC coated
Reinforcement	Multi-axial laid construction of galvanized steel wires
Weight	1350 g/m²
Heat- and cold resistance	-30°C to +70°C
Flame resistant	Not classified
Comments	Space for fabric folding must be increased by 100 mm on each side of the door, to avoid fabric wear.

Note! Must always be quoted by ASSA ABLOY Entrance Systems.

2.5.1.8 Vision panels

Application	Light admission and view through
Use	Only for standard and arctic fabric
Standard sizes	Width 800 or 1300 mm, height depending on door size
Material	Elaston 064, 1 mm
Weight	1230 g/m²
Hardness	77° shore acc. DIN 53505
Heat- and cold resistance	-30°C to +50°C
Tear resistance acc. DIN 53455	Along : 21 N/mm² Crosswise : 20 N/mm²

2.6 Operating system

2.6.1 General specifications

Control system	PLC-based
Protection class, control cabinet	IP65
Protection class, safety boxes	IP67
Protection class, motor/brake	IP55
Protection class, push buttons	IP65
Power supply	3/phase 400V 50Hz. Other alternatives available on request
Control voltage	24V DC
Fusing	20-25 A
Heat and cold resistance	-35 °C to +70 °C
Motor ratings	3.2 - 5.5 kW
Number of motors	One (two for big single leaf belt doors).

3. CEN Performance

The following tests have been carried out by the Swedish National Testing and Research Institute (SP) in Borås. For more detailed information and values, see ITT report: 0402-CDP-397307.

3.1 Lifetime expectation

- 100.000 door cycles

3.2 Resistance to windload

EN12424	
Test result	Class 3-5 (depending on door size).

Class	**Pressure Pa (N/m^2)**	**Specification**
0	-	No performance determined
1	300	
2	450	
3	700	
4	1000	
5	> 1000	Exceptional : Agreement between manufacturer and supplier

3.3 Resistance to water penetration

EN12425	
Test result	Class 3 (110 Pa)

Class	**Pressure Pa (N/m^2)**	**Specification**
0	-	No performance determined
1	30	Waterspray for 15 minutes
2	50	Waterspray for 20 minutes
3	> 50	Exceptional : Agreement between manufacturer and supplier

3.4 Air permeability

EN12426	
Test result	Class 2-3. Depending on size.

Class	**Air permeability dp at a pressure of 50 Pa (m^3/m^2/h)**
0	-
1	24
2	12
3	6
4	3
5	1,5
6	Exceptional : Agreement between manufacturer and supplier

3.5 Thermal transmittance

EN12428	
Thermal transmittance	Depending on door size. Specific data available on request.

3.6 Acoustic insulation

ISO 717	
Acoustic insulation	15 dB

3.7 Operating forces and safe openings

EN12453 & EN12604	**Crushing force N**	**Crushing force N**	**Crushing force N**
Opening gap mm	200 mm from lateral border right from outside	In the middle of the door opening	200 mm from lateral border left from outside
50 mm	passed	passed	passed
300 mm	passed	passed	passed

The crushing force is the force needed for the safety edge to be activated. The maximum force allowed, according to EN12453 safety in use of power operated doors is 400 N within a maximum period of time of 0.75s.

4. Building preparations

The Megadoor VL3190 vertical lifting fabric door is delivered for installation on site. To ensure efficient and quick fitting, the site must be prepared before the fitters arrive.

4.1 Installation

The doors can be easily adapted for several types of openings. The door leaf is compressed when opened and therefore takes up a minimum of space above the opening. The header box is screwed or welded. Alternatively it can be fixed with beam clamps to existing beams.

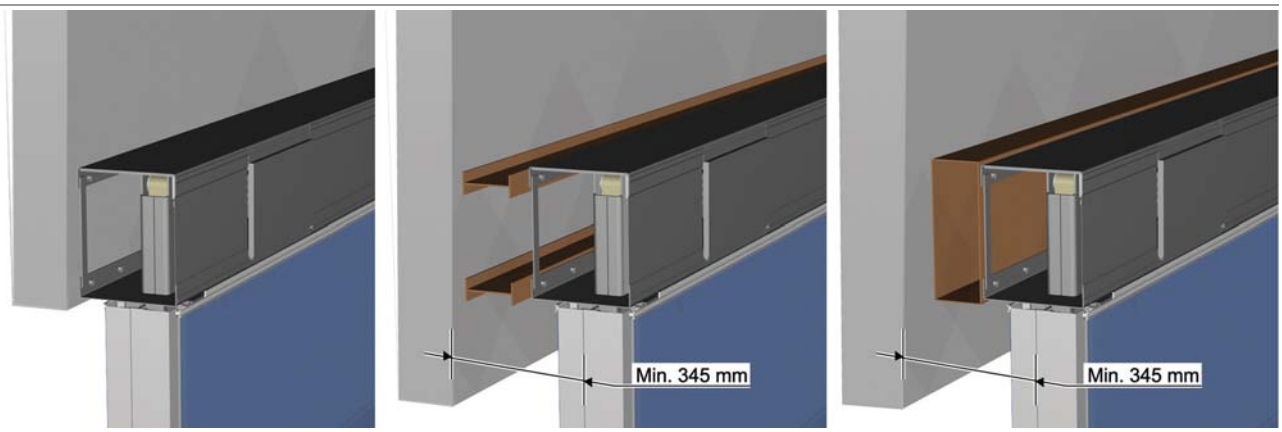
4.1.1 Installation of the header box

There are two basic methods for attaching vertical lifting fabric doors:

- Against a wall on the inside/outside of the opening
- In a door opening

4.1.1.1 Fitting against wall on inside/outside of opening

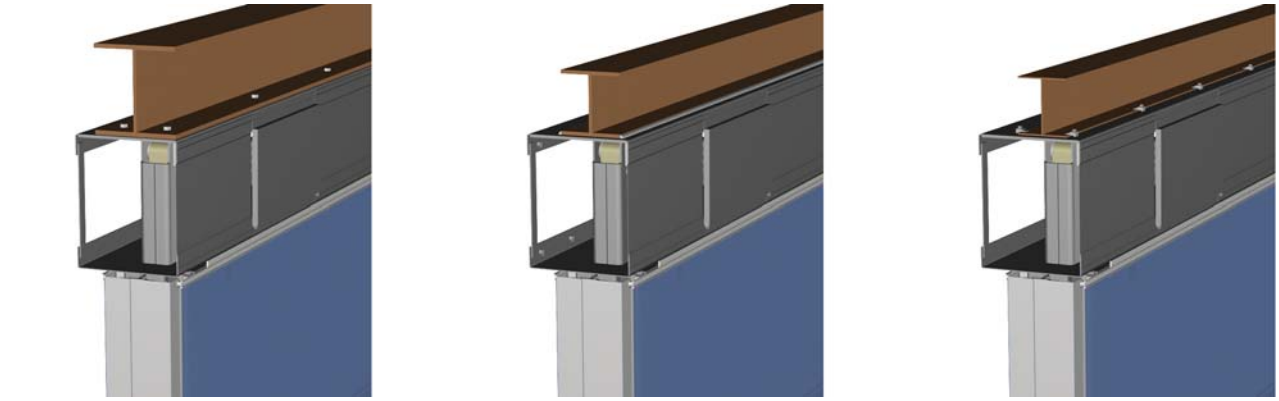
Internal mounting is recommended where there is available space. The drive machinery and guide rails will then be fully protected. Choose fitting on the outside of the opening if the environment in the building is harsh, or if there is insufficient space above the opening.



Fitting on wall against inside/outside of opening
N.B. Minimum 345 mm from wall to centre of guide rail. (85 mm from wall to rear side of header box).

4.1.1.2 Fitting in door opening

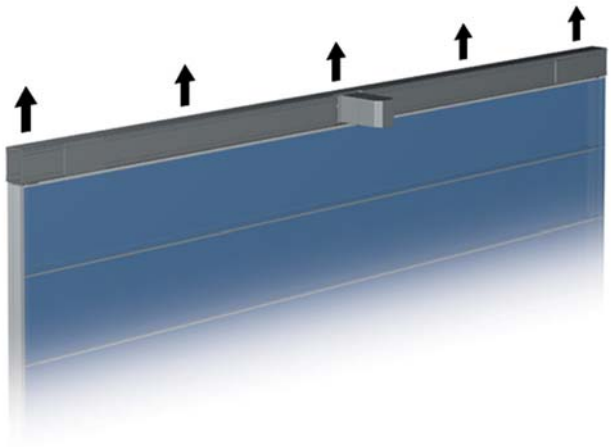
Mounting the door in an opening is an excellent alternative where the door is to be fitted in an existing opening and where the risk of colliding with the guide rails is negligible. It is also possible to protect the guide rails with a collision barrier.



Fitting in door opening (header box screwed, welded or fixed beam clamps).

4.1.1.3 Load on the building with door closed

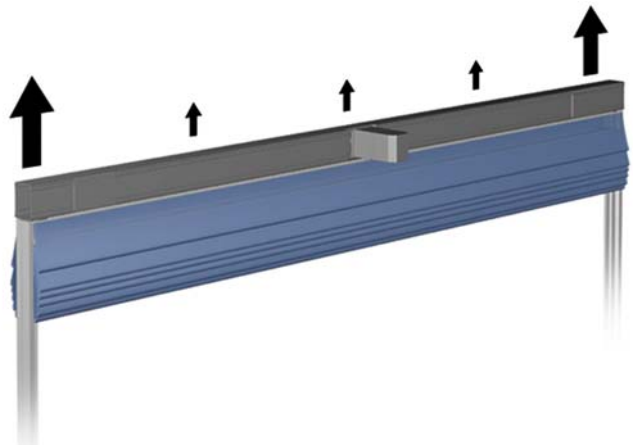
When the door is closed, the total weight is distributed on the fixing points. The distance between fixing points is about 1 m and must not exceed 2.5 m. Information on the total weight of the door will be provided at the time of quotation.



Load on building with door closed

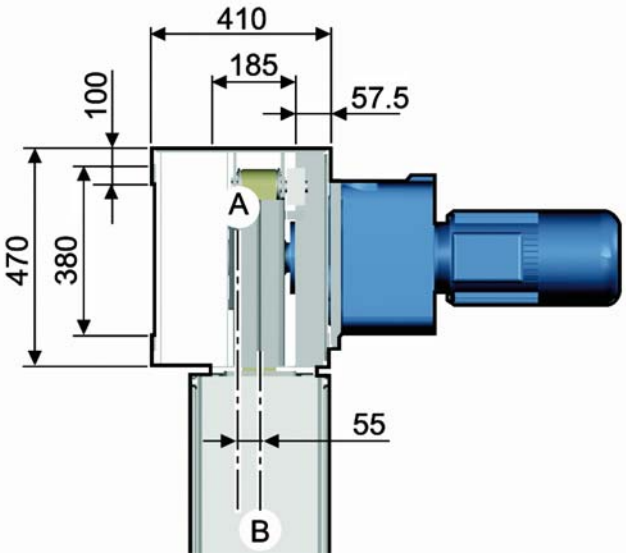
4.1.1.4 Load on the building with door open

The weight of the door is successively transferred to the ends of the header box as the door is opened. When the door is fully open, the door leaf weighs on the ends of the header box only. The weight of the header box itself continues to rest on all the fixing points.



Load on building with door open.

4.1.2 Mounting surface for header box



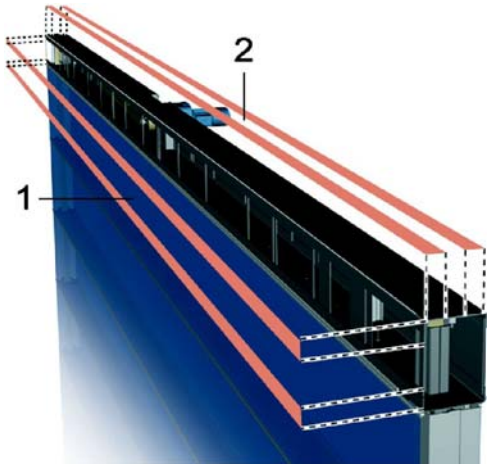
Screw holes in header box
A = Centre line header box
B = Centre line door leaf and guide rail

4.1.2.1 Mounting on wall (alt.1)

There must be flat, vertical surfaces to secure the header box (the part indicated as # 1).

4.1.2.2 Mount in opening (alt. 2)

There must be flat, horizontal surfaces to secure the header box (the part indicated as # 2).



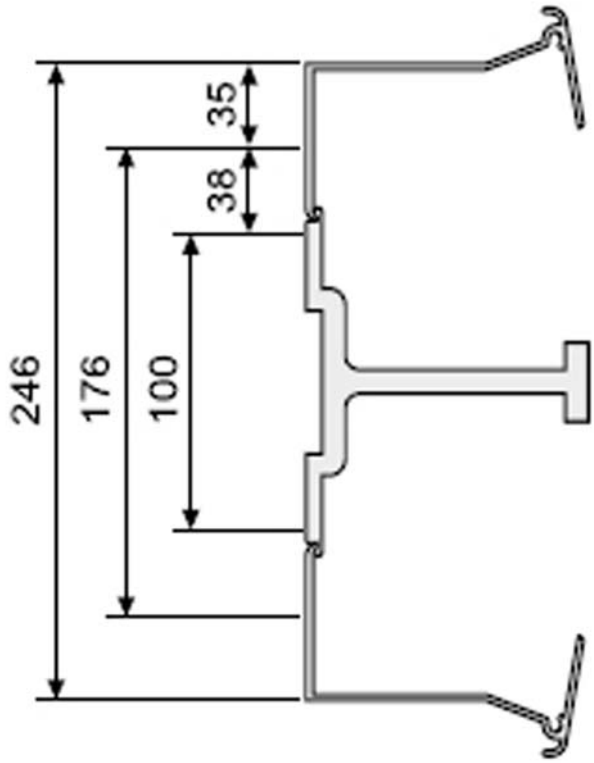
Mounting alternatives:

1) Fitting against wall
2) Fitting in opening

4.2 Mounting surface for guide rails

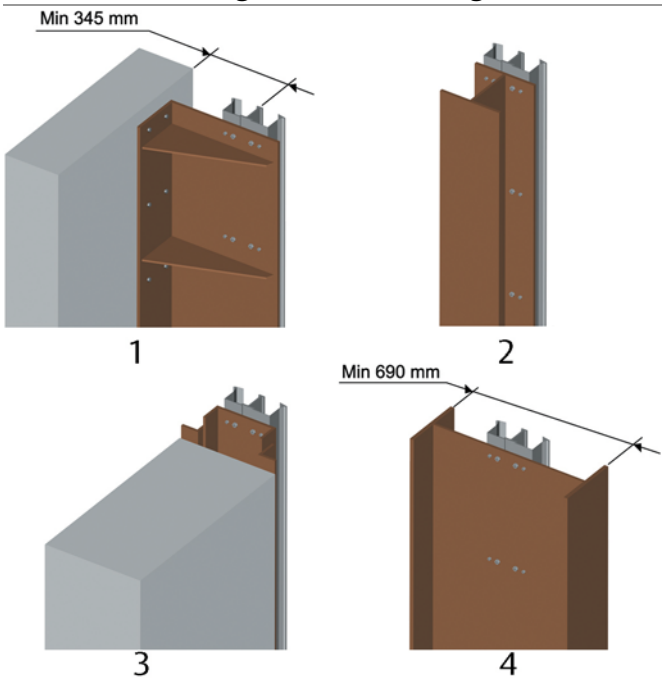
Suitable vertical surfaces are required on which to mount the guide rails. The mounting surfaces must be strong, flat and smooth. They must be parallel and deviate by no more than 5 mm from the vertical and by no more than 2 mm/m in the inward/outward direction from the vertical. The distance between fixing points is about 1 m.

N.B. Megadoor is not responsible for the calculation or supply of mounting surfaces, or for sealing between door and building.



Screw hole distances in guide rails

4.2.1 Mounting alternatives for guide rails



1. Fitting against wall.
2-4. Fitting in opening.

4.3 Installation of the guide rails

4.3.1 Installation of the control unit

The location of the control unit is best decided as follows :

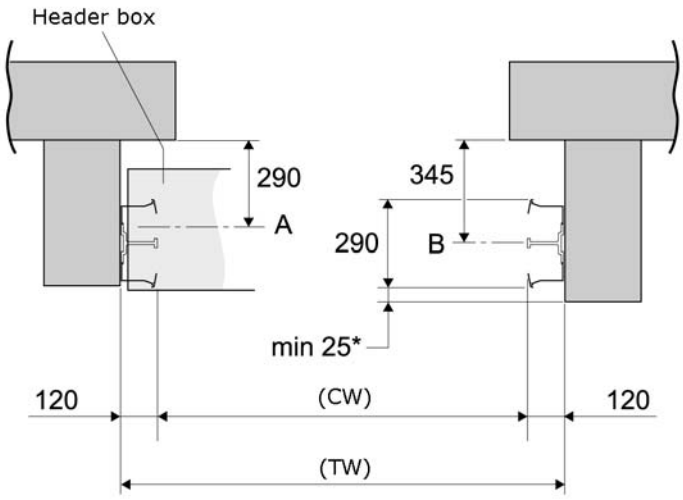
Environment	Effect on control unit	Location of control unit
Normal environment	Negligible effect, IP65 protection is sufficient.	Close to the door
Harsh interior environment	When opened for maintenance, dust and moisture may enter	In a safe area
Sustainable temperature difference inside/outside	Condensation when door is opened	Away from the door. Push button unit close to the door
Strongly corrosive environment, no safe location possible	Optimum protection required	Stainless steel control unit

Also consider the space requirements of the control unit.

5. Space requirements

TH	Total height	Distance between floor and top of header box
CH	Clear height	Distance between floor and bottom of door leaf when door is fully opened
OH	Over height	Vertical space required above the clear height
TS	Total space requirement	Distance between outer side of jambs
TW	Total width	Distance between the left and right vertical installation surfaces.
CW	Clear width	Clearance distance between the left and the right guide rails.
MD	Motor depth	Depth of the header box + gear motor + extra space for hand crank
A		Door leaf thickness
B		Minimum free space required for fabric folding
C		Distance from rear side of header box to guide rail centre

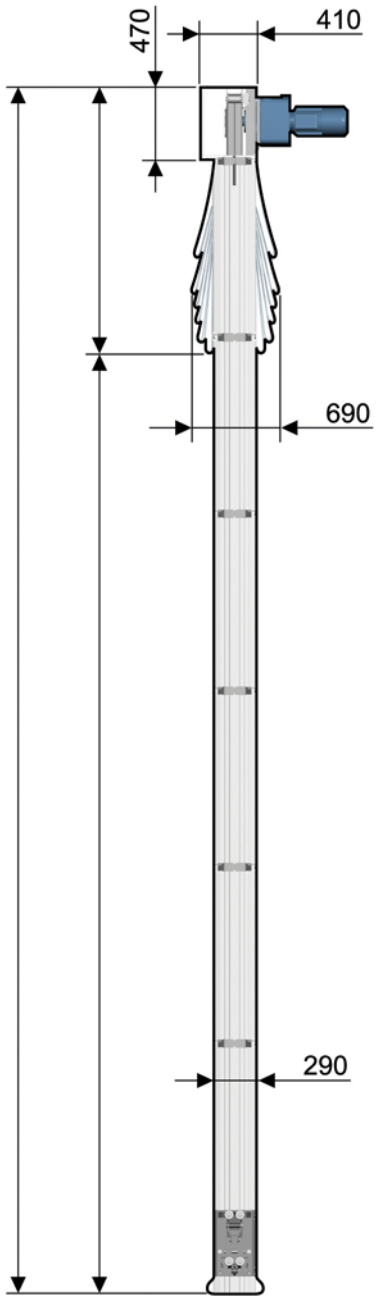
5.1 Space requirements for header box



*If outside of external wall
A = Centre line header box
B = Centre line door leaf and guide rail

5.2 Space requirements for operation

In contrast to other types of doors, the Megadoor VL3190 vertical lifting fabric door requires only limited top and side space. The door leaf is compressed when opened. Even for a large door, the requirements are minimal.



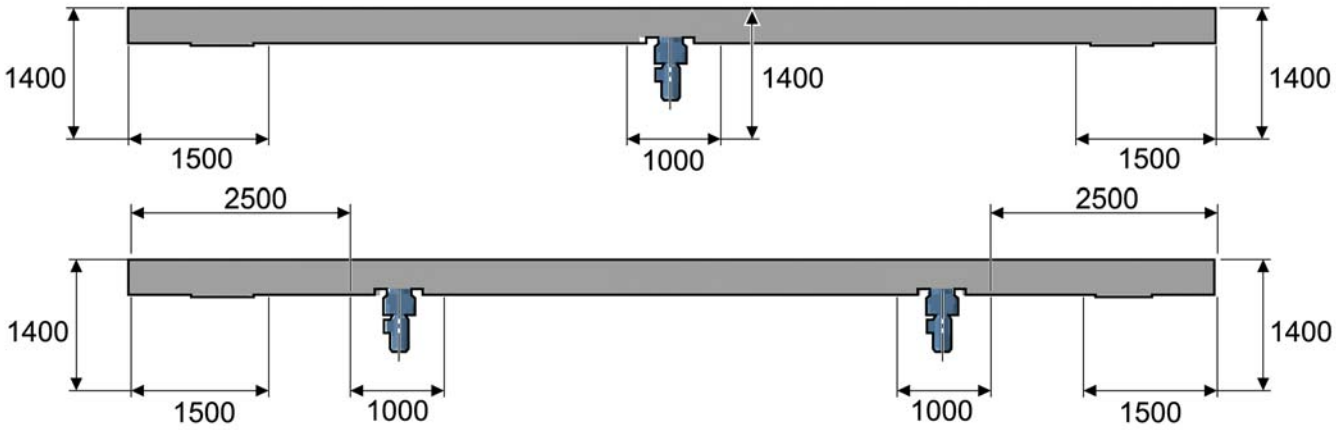
The difference in height caused by different widths, wind loads and motor types makes it impossible to stipulate simple formulas for height calculations. Contact Megadoor for information.

5.3 Space requirements for control unit

The following dimensions (w x h) may be of assistance in deciding where to place the control unit, brake resistor, possible additional cables or an additional safety switch for the power supply:

Control unit	700 x 700 mm

5.4 Space requirements for maintenance



Sizes in mm

6. Service

Preventive Maintenance Program and Modernization Services

As your entrances are part of your business flow, there's every reason to keep them working well. ASSA ABLOY Entrance Systems offers you a maintenance and modernization expertise to rely on. Our Maintenance Programs and Modernization Services are backed by a extensive expertise for all types of industrial door and docking systems, independent of brand. At your disposal is a team of dedicated expert technicians, proven through decades of maintenance, service and satisfied customers.

Preventive Maintenance Programs

Minimizing lost time, lost energy and unexpected hassle is our team's constant objective. Our service organization can support you 24/7 in maintaining all industrial door and docking systems, independent of brand. If you want to be one step ahead of break-downs, explore our portfolio of Pro-Active Care plans. Naturally, we also offer entrance upgrades to suit your specific wishes and business needs.

Pro-Active Care - Maintenance plans to fit your business

Regular maintenance can extend the lifetime of your equipment and help prevent unexpected problems. Our technician arrives on-site equipped with the knowledge and tools to service all automatic entrances, independent of brand.

- **Pro-Active Bronze**

The base on which all Pro-Active Plans are built provides the security of knowing that your equipment is regularly inspected and certified for safety, as well as performing optimally. It includes a number of planned on-site visits depending on your needs. Any unplanned service calls required during the term of the contract (including labor, travel and parts) are billed at special Pro-Active Care prices.

- **Pro-Active Silver**

This plan provides all the benefits of Pro-Active Bronze with the added advantage of labor and travel being included for service calls during regular business hours. The only additional charge would be for any parts that may be needed throughout the term of the contract.

- **Pro-Active Gold**

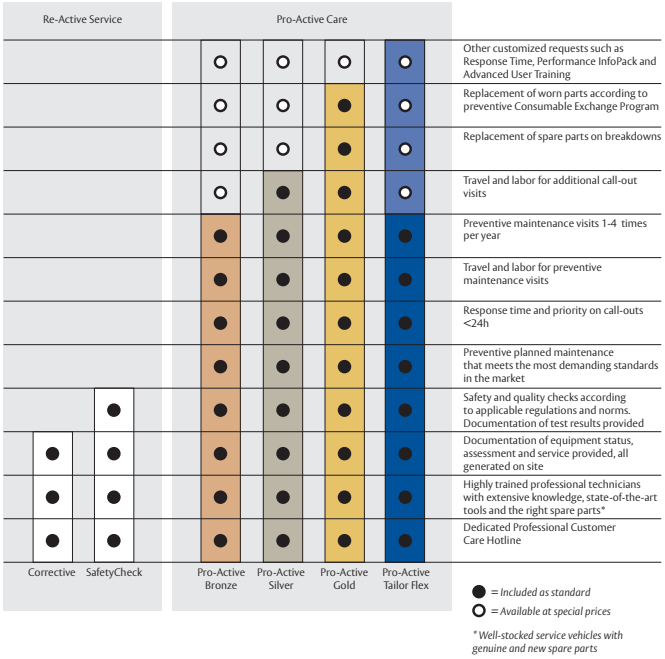
This plan provides the ultimate protection for your automatic entrance investment. It includes all the benefits of Pro-Active Silver, plus replacement of any parts required during an unplanned repair or planned maintenance visit. Pro-Active Gold is an excellent way to budget your automatic door expenses annually.

- **Pro-Active Tailor-Flex**

Our most flexible maintenance and service offering. The Pro-Active Care plan is designed by you, our customer. The plan allows you to balance your maintenance expenses against your real-world budget and presents the option to add or delete a number of maintenance elements to suit your budget goals, while meeting your overall performance and safety needs.

Modernization

Your entrances are a long-term investment, from which you always want the best. Products develop over time, so do regulations and your business. Let us help you increase energy savings and meet today's standards. We provide advice and modernization kits for outdated installations, ensuring your investment meet requirements and performs optimally for many more years to come.



Index

A

Access and automation 13
Acoustic insulation 21
Additional automatic functions 13
Air permeability 20
Arctic fabric 16

B

Bottom section 7
Building preparations 22

C

CEN Performance 20
Clear width and clear height 14
Colors 8
Construction 7
Control functions 13
Control unit 12
Copyright and Disclaimer Notice 2

D

Description 6
Door leaf 7, 15
Door leaf material 8
Drive unit 11

E

Electrical operation 11
Emergency power switch 13
Enclosed "non-motor side" (extra) .10
Enclosed motor side 10
Enclosing the header box 10
Environmental tolerance 14
External push button box 13

F

Fabric data 15
Features 3
Fitting against wall on inside/outside of opening 22
Fitting in door opening 22
Free contacts 13

G

General 6
General specifications 19
Guide rails 9

H

Header box 9
Heat resistant fabric - Aluminium coating 18
Heat resistant fabric - Aluminium polyurethane coating 18
Heat resistant fabric - Silicone coating 17

I

Installation 22
Installation of the control unit 25
Installation of the guide rails 25
Installation of the header box 22
Intermediate section 7

L

Lifetime expectation 20
Lifting belts 11
Load on the building with door closed 23
Load on the building with door open 23

M

Mount in opening (alt. 2) 23
Mounting alternatives for guide rails 24
Mounting on wall (alt.1) 23
Mounting surface for guide rails 24
Mounting surface for header box 23

O

Operating forces and safe openings 21
Operating system 11, 19
Optional colors 8
Options 6, 8

P

Performance 3
PLC 12
Protective casing over motor (extra) 10

R

Reduced opening 13
Resistance to water penetration 20
Resistance to windload 20

S

Safety arresters 7
Safety boxes 12
Safety functions 13
Safety photocells 1-channel 13
Security fabric 19
Service 29
Sound reduction fabric 17
Space requirements 26
Space requirements for control unit 28
Space requirements for header box 26
Space requirements for maintenance 28
Space requirements for operation .27
Specifications 14
Standard 6
Standard colors 8
Standard fabric 15
Surface treatment 14

T

Technical facts 3
Temperature control 12
Thermal transmittance 21

V

Vision panels 19

W

Warning lights - Orange flashing lights 13
Wind locking 7

ASSA ABLOY Entrance Systems is a leading supplier of entrance automation solutions for efficient flow of goods and people. With our globally recognized product brands Besam, Crawford, Megadoor and Albany, we offer products and services dedicated to satisfying end-user needs for safe, secure, convenient and sustainable operations.
ASSA ABLOY Entrance Systems is a division within ASSA ABLOY.

assaabloyentrance.com

ASSA ABLOY

Q1.0 - 2015

ASSA ABLOY Entrance Systems

assaabloyentrance.com

NIHVA